ITA06020 MACHINE LEARNING

Experiment 17 – Mobile Price Prediction Using Machine Learning

Aim:
To predict the price range of mobile phones based on their features using a machine learning classification algorithm.

Procedure:

1. Import the required Python libraries.
2. Load the Mobile Price Classification dataset.
3. Separate input features and target values.
4. Split the dataset into training and testing data.
5. Train a Random Forest classifier.
6. Predict the mobile price range and calculate accuracy.

Dataset: Mobile Price Classification Dataset (Kaggle). It contains mobile features such as RAM, battery power, internal memory and screen size.

Python Code:

```
import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import accuracy_score

url =
"https://raw.githubusercontent.com/ApnaClassroom/Dataset/main/MobilePriceRange.csv"

data = pd.read_csv(url)

X = data.drop("PriceRange", axis=1)

y = data["PriceRange"]

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)

model = RandomForestClassifier(random_state=42)
```

```
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", round(accuracy_score(y_test, y_pred) * 100, 2), "%")

sample = X_test.iloc[[0]]

prediction = model.predict(sample)

print("Predicted Price Range:", prediction[0])

print("Actual Price Range:", y_test.iloc[0])
```

Output:

```
Accuracy: 87.5 %
Predicted Price Range: Low
Actual Price Range: Low
```

Result:
The mobile price prediction model was successfully trained using the Random Forest algorithm to classify mobile phones into different price ranges.